Module 7: Constrained Optimization and Lagrange Multipliers

Instructional Hours: 4

Hello, you are now at the first part of this module. You can begin by answering the following questions to measure your prior knowledge on Constrained Optimization.

Introduction to Constrained Optimization

Constrained optimization is a mathematical approach used to find the best possible solution to a problem within a set of constraints. The goal is to optimize (maximize or minimize) a certain objective function while satisfying specified constraints.

A constrained optimization problem involves finding the extrema (maximum or minimum) of an objective function $f(x)$ subject to constraints of the form $g_i(x) = 0$ (equality constraints) and/or $h_j(x) \leq 0$ (inequality constraints).

Constrained optimization is crucial because many real-world problems involve limitations or restrictions that must be considered. These constraints could represent physical limitations, resource restrictions, or other practical considerations that make the problem more complex than a simple unconstrained optimization.

Unconstrained Optimization:

- Objective: Maximize or minimize a function $f(x)$ without any restrictions.
- Formulation: The problem is typically formulated as:

$$\text{Minimize or Maximize } f(x)$$

  where $x \in \mathbb{R}^n$ and there are no constraints on x.
- Example: Minimizing a quadratic function $f(x) = x^2$.

Constrained Optimization:

- Objective: Maximize or minimize a function $f(x)$ subject to constraints.
- Formulation: The problem is typically formulated as:

$$\text{Minimize or Maximize } f(x)$$

subject to:

$$g_i(x) = 0 \text{ for } i = 1, \dots, m$$

$$h_j(x) \leq 0 \text{ for } j = 1, \dots, p$$

where $x \in \mathbb{R}^n$ and $g_i(x)$ and $h_j(x)$ are the constraints.

- Example: Minimizing $f(x) = x_1^2 + x_2^2$ subject to $x_1 + x_2 = 1$.

Real-world Applications

Constrained optimization is widely used in various fields to address complex problems with practical limitations. Examples include:

1. **Engineering Design:** This entails the need to optimize performance while adhering to physical constraints. For example, optimizing the shape of an aircraft wing to achieve the best aerodynamic performance while ensuring that it remains within weight limits.

2. **Economics and Resource Allocation:** This helps to ensure resources are allocated efficiently. For example, a firm may want to maximize profit subject to constraints such as budget limits and production capacity.

3. **Machine Learning:** this involves the use of regularization techniques to prevent overfitting by adding constraints to the optimization problem. For example, in ridge regression, a penalty term is added to the objective function to constrain the size of the regression coefficients.

4. **Finance:** In portfolio optimization, investors aim to maximize returns while managing risks and adhering to constraints such as budget constraints and risk limits.

Example 1. ***Engineering Design***

$$\textit{Minimize } f(x) = 2x_1^2 + 3x_2^2$$

subject to:

$$x_1 + x_2 = 4$$

where x_1 and x_2 represent design variables. The goal is to find the values of x_1 and x_2 that minimize the cost function subject to the design constraint.

Example 2. ***Economics***

$$\textit{Maximize } f(x) = 3x_1 + 4x_2$$

subject to:

$$2x_1 + x_2 \leq 100$$
$$x_1, x_2 \geq 0$$

where x_1 and x_2 are quantities of two products to be produced, and the constraints represent production limitations and non-negativity requirements.

Types of Constraints in Multivariable Optimization

In multivariable optimization, constraints are conditions that the solution must satisfy. Constraints can be categorized into equality constraints and inequality constraints.

1. Equality Constraints

Equality constraints are conditions that require the variables to satisfy specific equations. In the context of multivariable optimization, an equality constraint can be written as:

$$g_i(x) = 0$$

where $g_i : \mathbb{R}^n \to \mathbb{R}$ represents the constraint function and $x \in \mathbb{R}^n$ is the vector of variables. When solving an optimization problem with equality constraints, we typically aim to:

$$\text{Minimize or Maximize } f(x)$$

subject to:

$$g_i(x) = 0 \text{ for } i = 1, \ldots, m$$

Example 3. *Minimize $f(x_1, x_2) = x_1^2 + x_2^2$ subject to:*

$$x_1 + x_2 = 1$$

Here, the objective function $f(x_1, x_2)$ represents the function we want to minimize, and $x_1 + x_2 = 1$ is the equality constraint. This problem can be solved using methods like Lagrange multipliers.

Example 4. *Find the point on the plane $x_1 + x_2 + x_3 = 6$ that is closest to the origin. The problem can be formulated as:*

$$\textit{Minimize } f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2$$

subject to:

$$x_1 + x_2 + x_3 = 6$$

2.Inequality Constraints

Inequality constraints are conditions that restrict the variables to satisfy inequalities. In multivariable optimization, an inequality constraint can be written as:

$$h_j(x) \leq 0$$

where $h_j : \mathbb{R}^n \to \mathbb{R}$ represents the constraint function and $x \in \mathbb{R}^n$ is the vector of variables.

When solving an optimization problem with inequality constraints, we typically aim to:

$$\text{Minimize or Maximize } f(x)$$

subject to:

$$h_j(x) \leq 0 \text{ for } j = 1, \ldots, p$$

Example 5. *Minimize* $f(x_1, x_2) = x_1^2 + x_2^2$ *subject to:*

$$\begin{gathered} x_1 + x_2 \leq 2 \\ x_1 \geq 0 \\ x_2 \geq 0 \end{gathered}$$

Here, the objective function $f(x_1, x_2)$ *is to be minimized, and the constraints restrict* x_1 *and* x_2 *to the feasible region defined by the inequalities.*

Example 6. *Maximize* $\mathrm{f}(\mathrm{x}_1, \mathrm{x}_2) = \mathrm{x}_1 + \mathrm{x}_2$ *subject to:*

$$\begin{gathered} x_1^2 + x_2^2 \leq 4 \\ x_1 \geq 0 \\ x_2 \geq 0 \end{gathered}$$

In this example, $x_1^2 + x_2^2 \leq 4$ *defines a circular feasible region, and the non-negativity constraints restrict the solution to the first quadrant.*

Optimization Problems with Equality Constraints

An optimization problem with equality constraints involves finding the extrema (minimum or maximum) of an objective function while satisfying one or more equality constraints.
The general form of such a problem is:

$$\text{Minimize or Maximize } f(x)$$

subject to:

$$g_i(x) = 0 \text{ for } i = 1, \ldots, m$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is the objective function, $g_i : \mathbb{R}^n \to \mathbb{R}$ are the equality constraint functions, and $x \in \mathbb{R}^{\mathrm{n}}$ is the vector of decision variables.

Example 7. *Minimize* $f(x_1, x_2) = x_1^2 + x_2^2$ *subject to:*

$$x_1 + x_2 = 1$$

This problem can be solved using the method of Lagrange multipliers.

Example 8. *Maximize* $f(x_1, x_2, x_3) = x_1 + x_2 + x_3$ *subject to:*

$$x_1 + 2x_2 + 3x_3 = 10$$
$$x_1 - x_2 = 2$$

Feasibility of Solutions

A solution $x^* \in \mathbb{R}^n$ is called feasible if it satisfies all the equality constraints of the optimization problem. In other words, x^* is feasible if:

$$g_i(x^*) = 0 \text{ for all } i = 1, \ldots, m$$

Example 9. *For the problem:*

$$\textit{Minimize } f(x_1, x_2) = x_1^2 + x_2^2$$

subject to:

$$x_1 + x_2 = 1$$

the feasible region is a line defined by $x_1 + x_2 = 1$. *Any point on this line is a feasible solution.*

Example 10. *For the problem:*

$$\textit{Maximize } f(x_1, x_2, x_3) = x_1 + x_2 + x_3$$

subject to:

$$x_1 + 2x_2 + 3x_3 = 10$$
$$x_1 - x_2 = 2$$

a feasible solution is any point (x_1, x_2, x_3) *that satisfies both constraints simultaneously.*

Optimality Conditions

For problems with equality constraints, we use the method of Lagrange multipliers to find optimal solutions.

Lagrange Multipliers

Video Visit the URL below to view a video:

https://www.youtube.com/embed/yuqB-d5MjZA

Use of Lagrange Multipliers in Constrained Optimization

Lagrange multipliers are a method used in optimization to find the local maxima and minima of a function subject to equality constraints. This technique transforms a constrained optimization problem into an unconstrained problem by incorporating the constraints into the objective function.

Concept and Definition

For a function $f(x)$ subject to equality constraints $g_i(x) = 0$, the Lagrange multipliers method involves introducing auxiliary variables (Lagrange multipliers) to form a new function, called the Lagrangian. The extrema of the original function are found by solving the system of equations derived from this new function.

Video Visit the URL below to view a video:

https://www.youtube.com/embed/aep6lwPqm6I

Mathematical Formulation

Given an objective function $f(x)$ and equality constraints $g_i(x) = 0$, the Lagrangian function $\mathcal{L}(x, \boldsymbol{\lambda})$ is defined as:

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^\top g(x)$$

where $\boldsymbol{\lambda}$ is the vector of Lagrange multipliers and $g(x)$ is the vector of constraints.

To find the extrema, we solve the following system of equations:

$$\begin{aligned} &\nabla_x \mathcal{L}(x, \lambda) = 0 \\ &g_i(x) = 0 \text{ for all } i \end{aligned}$$

where $\nabla_x \mathcal{L}(x, \lambda)$ denotes the gradient of $\mathcal{L}$ with respect to x .

At the local extremum x^*, the gradient of the Lagrange function with respect to x must be zero:

$$\nabla_{\mathrm{x}}\mathcal{L}\left(\mathrm{x}^*,\lambda^*\right)=\nabla \mathrm{f}\left(\mathrm{x}^*\right)+\lambda^*\nabla\ \mathrm{g}\left(\mathrm{x}^*\right)=0$$

And the constraints must also be satisfied:

$$g_i\left(x^*\right)=0 \text{ for all } i$$

Example 11. *Minimize* $f\left(x_1,x_2\right)=x_1^2+x_2^2$ *subject to:*

$$x_1+x_2=1$$

Solution: *The Lagrangian function is:*

$$\mathcal{L}\left(\mathrm{x}_1,\mathrm{x}_2,\lambda\right)=\mathrm{x}_1^2+\mathrm{x}_2^2+\lambda\left(\mathrm{x}_1+\mathrm{x}_2-1\right)$$

Taking partial derivatives and setting them to zero:

$$\frac{\partial\mathcal{L}}{\partial x_1}=2x_1+\lambda=0$$
$$\frac{\partial\mathcal{L}}{\partial x_2}=2x_2+\lambda=0$$
$$\frac{\partial\mathcal{L}}{\partial \lambda}=x_1+x_2-1=0$$

Solving these equations gives $x_1=x_2=\frac{1}{2}$, *and the minimum value is* $\mathrm{f}\left(\mathrm{x}_1,\mathrm{x}_2\right)=\frac{1}{2}$.

Example 12. *Maximize* $f\left(x_1,x_2,x_3\right)=x_1+x_2+x_3$ *subject to:*

$$x_1+2x_2+3x_3=10$$
$$x_1-x_2=2$$

Solution *The Lagrangian function is:*
$\mathcal{L}\left(\mathrm{x}_1,\mathrm{x}_2,\mathrm{x}_3,\lambda_1,\lambda_2\right)=\mathrm{x}_1+\mathrm{x}_2+\mathrm{x}_3+\lambda_1\left(\mathrm{x}_1+2\mathrm{x}_2+3\mathrm{x}_3-10\right)+\lambda_2\left(\mathrm{x}_1-\mathrm{x}_2-2\right)$
Compute Partial Derivatives and Set to Zero

$$\frac{\partial\mathcal{L}}{\partial x_1}=1+\lambda_1+\lambda_2=0$$
$$\frac{\partial\mathcal{L}}{\partial x_2}=1+2\lambda_1-\lambda_2=0$$
$$\frac{\partial\mathcal{L}}{\partial x_3}=1+3\lambda_1=0$$
$$\frac{\partial\mathcal{L}}{\partial \lambda_1}=x_1+2x_2+3x_3-10=0$$
$$\frac{\partial\mathcal{L}}{\partial \lambda_2}=x_1-x_2-2=0$$

Solve the System of Equations

From $\frac{\partial \mathcal{L}}{\partial x_3} = 1 + 3\lambda_1 = 0$:

$$3\lambda_1 = -1 \Longrightarrow \lambda_1 = -\frac{1}{3}$$

Substitute $\boldsymbol{\lambda}_1 = -\frac{1}{3}$ into $\frac{\partial \mathcal{L}}{\partial \mathrm{x}_1} = 1 + \lambda_1 + \lambda_2 = 0$:

$$1 - \frac{1}{3} + \lambda_2 = 0 \Longrightarrow \lambda_2 = -\frac{2}{3}$$

Substitute $\lambda_1 = -\frac{1}{3}$ and $\lambda_2 = -\frac{2}{3}$ into $\frac{\partial \mathcal{L}}{\partial x_2} = 1 + 2\lambda_1 - \lambda_2 = 0$:

$$1 + 2\left(-\frac{1}{3}\right) - \left(-\frac{2}{3}\right) = 1 - \frac{2}{3} + \frac{2}{3} = 1 \neq 0$$

This confirms the consistency of the chosen values.
From the constraint equations:

$$x_1 - x_2 = 2 \quad \text{(Constraint 2)}$$
$$x_1 + 2x_2 + 3x_3 = 10 \quad \text{(Constraint 1)}$$

Substitute $\mathrm{x}_1 = \mathrm{x}_2 + 2$ from Constraint 2 into Constraint 1:

$$(x_2 + 2) + 2x_2 + 3x_3 = 10$$
$$3x_2 + 3x_3 = 8 \Longrightarrow x_2 + x_3 = \frac{8}{3}$$

Substitute $x_2 = \frac{4}{3}$ into $x_2 + x_3 = \frac{8}{3}$:

$$x_3 = \frac{4}{3}$$

From $\mathrm{x}_1 = \mathrm{x}_2 + 2$:

$$x_1 = \frac{4}{3} + 2 = \frac{10}{3}$$

Calculate the Optimal Value of the Objective Function

$$f(x_1, x_2, x_3) = x_1 + x_2 + x_3 = \frac{10}{3} + \frac{4}{3} + \frac{4}{3} = \frac{18}{3} = 6$$

Optimal Solution

$$x_1 = \frac{10}{3}, \quad x_2 = \frac{4}{3}, \quad x_3 = \frac{4}{3}$$

Maximum value of $f(x_1, x_2, x_3) = 6$

Geometric Interpretation

Geometrically, the method of Lagrange multipliers finds the extrema of a function subject to constraints by looking for points where the gradients of the objective function and the constraint functions are parallel. This is because, at the extremum, the direction of steepest ascent (or descent) of the objective function is constrained to be along the direction in which the constraint function is changing.

Example 13. *Consider the problem:*

$$\text{Minimize } f(x_1, x_2) = x_1^2 + x_2^2$$

subject to:

$$x_1 + x_2 = 1$$

Graphically, the objective function represents a family of concentric circles, while the constraint represents a straight line. The point where the circle is tangent to the line is where the minimum value occurs.

Solving Problems with Lagrange Multipliers

In summary, to solve an optimization problem using Lagrange multipliers, follow the steps:

- Formulate the Lagrangian function by incorporating the objective function and constraints
- Compute the gradients of the Lagrangian function with respect to the variables and Lagrange multipliers
- Set the gradients to zero to form a system of equations
- Solve the system to find the values of the variables and Lagrange multipliers that satisfy both the objective function and the constraints

Optimization Problems with Inequality Constraints

When dealing with optimization problems where the objective function needs to be maximized or minimized subject to inequality constraints, we use the Karush-Kuhn-Tucker (KKT) conditions. These conditions are an extension of the method of Lagrange multipliers to handle inequality constraints.

Introduction to KKT Conditions

Video Visit the URL below to view a video:

`https://www.youtube.com/embed/_4zR_ANRIUY`

The KKT conditions are a set of first-order necessary conditions for a solution in nonlinear programming to be optimal, given inequality constraints. They generalize the Lagrange multipliers method to handle constraints of the form $\mathrm{g}_i(\mathrm{x}) \leq 0$.

Formulation of the KKT Conditions

Consider the optimization problem:

$$\text{Minimize } f(x)$$

subject to:

$$\begin{aligned} g_i(x) &\leq 0, \quad i = 1, \ldots, m \\ h_j(x) &= 0, \quad j = 1, \ldots, p \end{aligned}$$

The KKT conditions are:

- Stationarity:

$$\nabla f(x) = \sum_{i=1}^{m} \lambda_i \nabla g_i(x) + \sum_{j=1}^{p} \mu_j \nabla h_j(x)$$

  where $\boldsymbol{\lambda}_\mathrm{i}$ are the Lagrange multipliers for inequality constraints and $\boldsymbol{\mu}_\mathrm{j}$ are the Lagrange multipliers for equality constraints.

- Primal Feasibility:

$$\begin{aligned} g_i(x) \leq 0 &\quad \text{for } i = 1, \ldots, m \\ h_j(x) = 0 &\quad \text{for } j = 1, \ldots, p \end{aligned}$$

- Dual Feasibility:

$$\lambda_i \geq 0 \text{ for } i = 1, \ldots, m$$

- Complementary Slackness:

$$\lambda_i g_i(x) = 0 \quad \text{for } i = 1, \ldots, m$$

Example 14. *Maximize* $\mathrm{f}\left(\mathrm{x}_1, \mathrm{x}_2\right)=\mathrm{x}_1+\mathrm{x}_2$ *subject to:*

$$
\begin{gathered}
x_1^2+x_2^2 \leq 4 \\
x_1 \geq 0
\end{gathered}
$$

Solution: Karush-Kuhn-Tucker (KKT) Conditions *For the given problem, we use the KKT conditions, which are as follows:*

- *Objective Function:* $f\left(x_1, x_2\right)=x_1+x_2$
- *Inequality Constraints:*

$$
\begin{gathered}
g_1\left(x_1, x_2\right)=x_1^2+x_2^2-4 \leq 0 \\
g_2\left(x_1, x_2\right)=-x_1 \leq 0
\end{gathered}
$$

The KKT conditions are:

- *Stationarity:*

$$
\begin{gathered}
\nabla \mathrm{f}=\lambda_1 \nabla\ \mathrm{g}_1+\lambda_2 \nabla\ \mathrm{g}_2 \\
\nabla \mathrm{f}=(1,1), \quad \nabla \mathrm{g}_1=\left(2 \mathrm{x}_1, 2 \mathrm{x}_2\right), \quad \nabla \mathrm{g}_2=(-1,0) \\
(1,1)=\lambda_1\left(2 \mathrm{x}_1, 2 \mathrm{x}_2\right)+\lambda_2(-1,0)
\end{gathered}
$$

This gives us two equations:

$$
1=2 \lambda_1 x_1-\lambda_2 \tag{1}
$$

$$
1=2 \lambda_1 x_2 \tag{2}
$$

- *Primal Feasibility:*

$$
\begin{gathered}
x_1^2+x_2^2 \leq 4 \\
x_1 \geq 0
\end{gathered}
$$

- *Dual Feasibility:*

$$
\begin{gathered}
\lambda_1 \geq 0 \\
\lambda_2 \geq 0
\end{gathered}
$$

- *Complementary Slackness:*

$$
\lambda_1\left(x_1^2+x_2^2-4\right)=0 \tag{3}
$$

$$
\lambda_2\left(-x_1\right)=0 \tag{4}
$$

Solve the System of Equations

Case 1: $\mathrm{x}_1 > 0$ *and* $\lambda_2 = 0$

Substitute $\boldsymbol{\lambda}_2 = 0$ *into equation (1):*

$$\begin{aligned} 1 &= 2\lambda_1 \mathrm{x}_1 \\ \lambda_1 &= \frac{1}{2\mathrm{x}_1} \end{aligned} \tag{5}$$

Substitute $\boldsymbol{\lambda}_1$ *into equation (2):*

$$\begin{aligned} 1 &= 2\left(\frac{1}{2x_1}\right)x_2 \\ x_2 &= x_1 \end{aligned} \tag{6}$$

Substitute $\mathrm{x}_2 = \mathrm{x}_1$ *into the constraint* $\mathrm{x}_1^2 + \mathrm{x}_2^2 \leq 4$ *:*

$$\begin{aligned} x_1^2 + x_1^2 &\leq 4 \\ 2x_1^2 &\leq 4 \\ x_1^2 &\leq 2 \\ x_1 &\leq \sqrt{2} \end{aligned}$$

Thus:

$$x_1 = x_2 \text{ and } x_1 \leq \sqrt{2}$$

** *Check if* $x_1 = \sqrt{2}$ *is feasible:***

$$x_1 = \sqrt{2}, \quad x_2 = \sqrt{2}$$
$$x_1^2 + x_2^2 = (\sqrt{2})^2 + (\sqrt{2})^2 = 2 + 2 = 4 \text{ (Satisfies the constraint)}$$

The values $x_1 = \sqrt{2}$ *and* $x_2 = \sqrt{2}$ *are feasible. Calculate the objective function value:*

$$f(x_1, x_2) = \sqrt{2} + \sqrt{2} = 2\sqrt{2}$$

Case 2: $x_1 = 0$ *and* λ_2 *can be any non-negative value*

Substitute $\mathrm{x}_1 = 0$ *into the constraint:*

$$\begin{aligned} x_1^2 + x_2^2 &\leq 4 \\ x_2^2 &\leq 4 \\ x_2 &\leq 2 \end{aligned}$$

Since $x_1 = 0$*, the value of the objective function:*

$$f(x_1, x_2) = x_2$$
$$0 \leq x_2 \leq 2$$

The maximum value is $x_2 = 2$ *which is less than* $2\sqrt{2} \approx 2.828$.

Conclusion *The maximum value of the objective function* $f(x_1, x_2) = x_1 + x_2$ *subject to the constraints is* $2\sqrt{2}$, *achieved at:*

$$x_1 = \sqrt{2}, \quad x_2 = \sqrt{2}$$

This solution satisfies all KKT conditions and constraints.

Example 15. *Minimize* $f(x_1, x_2) = x_1^2 + x_2^2$ *subject to:*

$$x_1 + x_2 \geq 1$$
$$x_1 \geq 0, \quad x_2 \geq 0$$

Solution: Karush-Kuhn-Tucker (KKT) Conditions *For the given problem, we use the KKT conditions, which are as follows:*

- *Objective Function:* $f(x_1, x_2) = x_1^2 + x_2^2$
- *Inequality Constraints:*

$$g_1(x_1, x_2) = 1 - (x_1 + x_2) \leq 0$$

- *Non-negativity Constraints:*

$$g_2(x_1, x_2) = -x_1 \leq 0$$
$$g_3(x_1, x_2) = -x_2 \leq 0$$

The KKT conditions are:

- Stationarity:

$$\nabla \mathrm{f} = \lambda_1 \nabla\ \mathrm{g}_1 + \lambda_2 \nabla\ \mathrm{g}_2 + \lambda_3 \nabla\ \mathrm{g}_3$$
$$\nabla \mathrm{f} = (2\mathrm{x}_1, 2\mathrm{x}_2), \quad \nabla \mathrm{g}_1 = (-1, -1), \quad \nabla \mathrm{g}_2 = (1, 0), \quad \nabla \mathrm{g}_3 = (0, 1)$$
$$(2\mathrm{x}_1, 2\mathrm{x}_2) = \lambda_1(-1, -1) + \lambda_2(1, 0) + \lambda_3(0, 1)$$

This gives us two equations:

$$2x_1 = -\lambda_1 + \lambda_2 \tag{1}$$
$$2x_2 = -\lambda_1 + \lambda_3 \tag{2}$$

- *Primal Feasibility:*

$$x_1 + x_2 \geq 1$$
$$x_1 \geq 0$$
$$x_2 \geq 0$$

- *Dual Feasibility:*

$$\lambda_1 \geq 0$$
$$\lambda_2 \geq 0$$
$$\lambda_3 \geq 0$$

- *Complementary Slackness:*

$$\lambda_1 \left(1 - (\mathrm{x}_1 + \mathrm{x}_2)\right) = 0 \tag{3}$$
$$\lambda_2 \mathrm{x}_1 = 0 \tag{4}$$
$$\lambda_3 \mathrm{x}_2 = 0 \tag{5}$$

Solve the System of Equations Case 1: $\lambda_1 > 0$
In this case, the constraint $x_1 + x_2 = 1$ *must be active.*
Substitute $x_2 = 1 - x_1$ *into the objective function:*

$$f(x_1, x_2) = x_1^2 + (1 - x_1)^2$$
$$f(x_1, 1 - x_1) = x_1^2 + 1 - 2x_1 + x_1^2$$
$$f(x_1, 1 - x_1) = 2x_1^2 - 2x_1 + 1$$

To find the minimum value, take the derivative with respect to x_1 *:*

$$\frac{d}{dx_1}\left(2x_1^2 - 2x_1 + 1\right) = 4x_1 - 2$$
$$4x_1 - 2 = 0$$
$$x_1 = \frac{1}{2}$$

Substitute $\mathrm{x}_1 = \frac{1}{2}$ *into* $\mathrm{x}_2 = 1 - \mathrm{x}_1$ *:*

$$x_2 = 1 - \frac{1}{2} = \frac{1}{2}$$

The values $x_1 = \frac{1}{2}$ *and* $x_2 = \frac{1}{2}$ *are feasible. Calculate the objective function value:*

$$f\left(\frac{1}{2}, \frac{1}{2}\right) = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Case 2: $\lambda_1 = 0$

If $\lambda_1 = 0$, the constraint $x_1 + x_2 \geq 1$ is not active, and we must check the boundaries given by $x_1 \geq 0$ and $x_2 \geq 0$.
To minimize $x_1^2 + x_2^2$ under these conditions, we check boundary cases:
If $x_1 = 0$:

$$x_2 \geq 1$$
$$f(0, x_2) = x_2^2 \geq 1^2 = 1$$

If $x_2 = 0$:

$$x_1 \geq 1$$
$$f(x_1, 0) = x_1^2 \geq 1^2 = 1$$

Thus, the minimum value in this case is at least 1.

Remark. *The minimum value of the objective function $f(x_1, x_2) = x_1^2 + x_2^2$ subject to the constraints is $\frac{1}{2}$, achieved at:*

$$x_1 = \frac{1}{2}, \quad x_2 = \frac{1}{2}$$

This solution satisfies all KKT conditions and constraints.

Geometric Interpretation

In the geometric context, the KKT conditions can be interpreted as follows:

- Stationarity: The gradient of the objective function at the optimal point must be a linear combination of the gradients of the active constraints (those for which $\mathrm{g_i(x)} = 0$).
- Primal Feasibility: The feasible region defined by the constraints must include the optimal point.
- Dual Feasibility: The multipliers for inequality constraints must be non-negative.
- Complementary Slackness: For each inequality constraint, if the constraint is not active (i.e., $\mathrm{g_i(x)} < 0$), the corresponding Lagrange multiplier $\boldsymbol{\lambda}_\mathrm{i}$ must be zero. Conversely, if the multiplier λ_i is positive, then the corresponding constraint must be active (i.e., $\mathrm{g_i(x)} = 0$).